UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

© Hak Cipta Universiti Teknologi MARA

PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	A	Which operator sends data to cout?	The insertion operator << writes to an output stream.
A2	D	Which keyword prevents a variable's value from being changed after initialisation?	const defines a read-only variable.
A3	A	Which phase of the program development life cycle defines what the program must do?	Analysis establishes the requirements before design and coding.
A4	D	Which part of a for loop is executed after each iteration?	The update expression runs after each iteration.
A5	A	What does a compiler do?	A compiler translates the complete source file into object code.
A6	C	Which structure chooses between two alternative paths?	if...else performs two-way selection.
A7	D	Which data type stores a whole number?	int stores whole numbers; float stores real values.
A8	C	Which loop is guaranteed to execute its body at least once?	do...while tests the condition after the body.
A9	A	What are parallel arrays?	Related data is stored at the same index in each array.
A10	D	Which values can a bool variable hold?	A bool holds true or false.

© Hak Cipta Universiti Teknologi MARA

PART B — WORKED ANSWERS

QUESTION 1

Topic: debugging testing

a) Define source code and object code.

Source code is the human-readable program text; object code is the machine-oriented output produced by a compiler.

Marking: Award 1 mark for each definition.

(3 marks)

b) Describe THREE stages of the program development life cycle.

Analysis (requirements), design (algorithm/flowchart), coding (implementation), testing (verification), maintenance (updates). Any three stages.

Marking: Award 1 mark for each correctly described stage.

(2 marks)

QUESTION 2

Topic: control structures

a) Draw a flowchart for the sum and average calculation.

Start -> read N -> total = 0, count = 1 -> while count <= N add count -> average = total / N -> display total and average -> end.

Marking: Award marks for input, initialisation, loop condition, accumulator, average and output.

(8 marks)

b) Write pseudocode and a complete C++ program for the algorithm.

Use a for loop from 1 to N, accumulate total, compute average as double(total)/N, then display both values.

Marking: Award marks for loop, accumulator, correct average type and output.

(5 marks)

QUESTION 3

Topic: data types

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

int age; double price; const double DISCOUNT = 0.10; char grade;

Marking: Award marks for correct types, names and const initialisation.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

cin >> age >> price; price = price - price * DISCOUNT; cout << price;

Marking: Award marks for input, discount expression and output.

(5 marks)

QUESTION 4

Topic: control structures

a) Trace the program and state the output.

30

Marking: The sum of 2+4+6+8+10 is 30.

(7 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

int i=0,s=0; do { if(a[i]>5) s+=a[i]; i++; } while(i<5); cout<<s;

Marking: Award marks for do...while, condition, filter and output.

(8 marks)

© Hak Cipta Universiti Teknologi MARA

QUESTION 5

Topic: arrays

a) Declare an integer array of ten marks and read the marks from the user.

int marks[10]; for(int i=0;i<10;i++) cin >> marks[i];

Marking: Award marks for array declaration, loop and input.

(8 marks)

b) Write a program segment to calculate and display the average, highest mark and number of marks above the average.

Compute total and highest in one loop; average = double(total)/10; then second loop counts marks > average.

Marking: Award marks for total, average, highest, second loop and output.

(8 marks)

© Hak Cipta Universiti Teknologi MARA

PART C — WORKED ANSWERS

QUESTION 1

A catering company charges RM12.50 per meal. Orders of 50-99 meals cost RM11.50 each, 100-199 cost RM10.50 each, and 200 or more cost RM9.50 each.

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;
int main() {
    string name;
    int quantity, grandQuantity = 0;
    double unitPrice, total, grandTotal = 0;
    char again;
    do {
        cout << "Customer name: ";
        getline(cin >> ws, name);
        cout << "Quantity: ";
        cin >> quantity;
        if (quantity < 0) {
            cout << "Invalid quantity" << endl;
        } else {
            if (quantity >= 200) unitPrice = 9.50;
            else if (quantity >= 100) unitPrice = 10.50;
            else if (quantity >= 50) unitPrice = 11.50;
            else unitPrice = 12.50;
            total = quantity * unitPrice;
            grandTotal += total;
            grandQuantity += quantity;
            cout << fixed << setprecision(2);
            cout << "Receipt for " << name << ": "
                 << quantity << " x RM" << unitPrice
                 << " = RM" << total << endl;
        }
        cout << "Another customer? (y/n): ";
        cin >> again;
    } while (again == 'y' || again == 'Y');
    cout << "Total quantity: " << grandQuantity << endl;
    cout << "Total sales: RM" << fixed << setprecision(2)
         << grandTotal << endl;
    return 0;
}
```

Marking: Award 3 marks for input/validation, 6 for price selection and total, 3 for repetition, 2 for final totals and 1 for output clarity.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

```
#include <iostream>
#include <string>
using namespace std;
struct Book {
    string title;
    double price;
    int quantity;
};
int main() {
    Book books[5];
    double totalValue = 0;
```

© Hak Cipta Universiti Teknologi MARA

```
    int mostExpensive = 0;
    for (int i = 0; i < 5; i++) {
        cin.ignore();
        cout << "Title: ";
        getline(cin, books[i].title);
        cout << "Price and quantity: ";
        cin >> books[i].price >> books[i].quantity;
        totalValue += books[i].price * books[i].quantity;
        if (books[i].price > books[mostExpensive].price) {
            mostExpensive = i;
        }
    }
    cout << "Total inventory value: RM" << totalValue << endl;
    cout << "Most expensive: " << books[mostExpensive].title
         << " (RM" << books[mostExpensive].price << ")" << endl;
    return 0;
}
```

Marking: Award 3 marks for struct definition, 4 for input, 4 for inventory value, 3 for most expensive and 1 for output.

(15 marks)

END OF MARKING SCHEME

© Hak Cipta Universiti Teknologi MARA